

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Duration : 1 Hour

Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I Shrimathi V (192571078) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Analytical Problem Solving

1. Water Jug Problem

Given:

- One **4-gallon jug**
- One **3-gallon jug**
- Unlimited water supply

Allowed operations:

- Fill either jug.
- Empty either jug.
- Pour water from one jug into another until one becomes empty or the other becomes full.

Goal: Obtain **exactly 2 gallons** in the **4-gallon jug**.

Method Selection:

This is a **State Space Search Problem**.

Each state is represented as:

(Water in 4-gallon jug, Water in 3-gallon jug)

Example:

(4,0)

means

- 4-gallon jug contains 4 gallons
- 3-gallon jug contains 0 gallons

Step	Action	State (4G,3G)
Initial	Empty jugs	(0,0)
1	Fill 3-gallon jug	(0,3)
2	Pour into 4-gallon jug	(3,0)
3	Fill 3-gallon jug again	(3,3)
4	Pour into 4-gallon until full	(4,2)
5	Empty 4-gallon jug	(0,2)
6	Pour remaining 2 gallons into 4-gallon	(2,0)

Final Answer:

The 4-gallon jug contains exactly 2 gallons.

Interpretation:

The Water Jug problem demonstrates how **state-space search** systematically explores possible states until the goal of state is reached.

2. Mars Rover Intelligent Agent

(i) Percepts

The rover receives information through:

- Cameras
- Temperature sensors
- Soil composition sensors
- Rock sample analyzers
- GPS/Navigation sensors
- Obstacle sensors
- Atmospheric sensors
- Battery level sensor

Property	Type
Observability	Partially Observable
Agents	Single Agent
Determinism	Stochastic
Episodes	Sequential
Dynamics	Dynamic
State Space	Continuous

(iii) Actions

The rover can:

- Move forward/backward
- Turn left/right
- Avoid obstacles
- Drill rocks
- Collect soil samples
- Capture images
- Analyze samples
- Transmit data
- Recharge using solar panels

iv) Performance Measure

Performance is evaluated using:

- Accuracy of collected samples
- Distance covered
- Energy efficiency
- Successful obstacle avoidance
- Scientific discoveries
- Communication success

(v) Suitable Agent Architecture

A **Utility-Based Learning Agent** is most suitable.

Reason:

- Makes decisions under uncertainty.
- Learns from previous experiences.
- Chooses actions that maximize mission success while minimizing energy consumption.

Interpretation

A Mars Rover must continuously perceive its environment, reason for uncertain situations, and choose the most beneficial action, making a utility-based learning architecture appropriate.

3. 8-Queens Problem Formulation

Problem Understanding

Place **8 queens** on an **8 × 8 chessboard** such that no queen attacks another.

Queens attack:

- Horizontally
- Vertically
- Diagonally

Method Selection

The problem is formulated as a **Search Problem**.

Problem Components

Initial State - Empty chessboard

State Space - All possible placements of queens.

Operators - Place one queen in a valid row of the next column.

Goal Test

Every queen satisfies:

- No same row
- No same column
- No same diagonal

Path Cost

Each placement has an equal cost.

Solution is found when all eight queens are safely placed.

Interpretation

The 8-Queens problem illustrates how AI uses search and constraint satisfaction techniques to efficiently find valid configurations.